2.17 Consider more complex Lagrangian Densities:

a) Let $\mathcal{L} = -\frac{1}{4\mu_0} D(F^{\mu\nu} F_{\mu\nu}) - k A_\mu j^\mu$ w/ $D(x)$

$$S = \frac{1}{c}\int [-\frac{1}{4\mu_0} D(F^{\mu\nu} F_{\mu\nu}) - k A_\mu j^\mu] \, d^4 x^\sigma$$

$$\delta S = \frac{1}{c}\int_a^b [-\frac{1}{4\mu_0} D(F^{\mu\nu} 2 \partial_\nu \delta A_\mu) - k j^\mu \delta A_\mu] d^4 x^\sigma$$

Integrate by parts to simplify the first term

$$D F^{\mu\nu} \partial_\nu \delta A_\mu = \partial_\nu (D F^{\mu\nu} \delta A_\mu) - \delta A_\mu \partial_\nu (D F^{\mu\nu})$$

$$\partial_\nu (D F^{\mu\nu}) = \partial_\nu D (F^{\mu\nu}) + D \partial_\nu F^{\mu\nu})$$

$$F^{\mu\nu} F_{\mu\nu} = 2(B^2 - \frac{E^2}{c^2})$$

b) $\partial_\alpha [D'(F^{\mu\nu} F_{\mu\nu}) F^{\alpha\beta}] = \mu_0 k j^\beta$

Find $\vec{E}$

$$D'(F^{\mu\nu} F_{\mu\nu}) F^{\alpha\beta} = \int \mu_0 k j^\beta \, d^4 x^\alpha$$

$$F^{\alpha\beta} = \int \frac{\mu_0 k j^\beta}{D'(F^{\mu\nu} F_{\mu\nu})} d^4 x^\alpha \qquad j^\beta = (cq, 0, 0, 0)$$

$$\frac{\vec{E}}{c} =$$

c) Born-Infeld theory (currently in vogue in String theory)
Born M and Infeld L, Phys. Rev. S A144, p. 425 (1934)